
The Digestive System

Core Functions



Ingestion & Digestion

Breakdown of food



Absorption

Absorbing them into the bloodstream.



Defecation

Eliminating indigestible waste materials from the body.

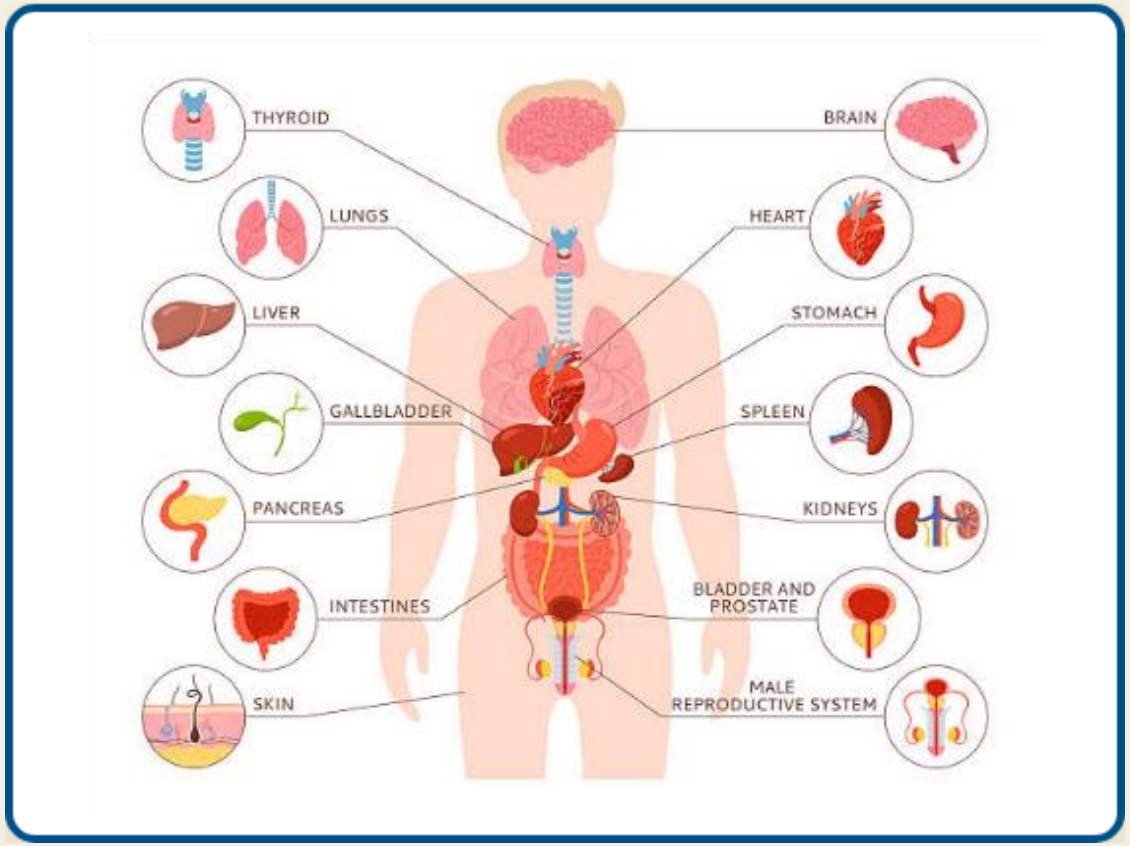
System Anatomy

Main Organs (GI Tract)

- **Mouth & Pharynx:** Entry and initial routing.
- **Esophagus:** Transport tube.
- **Stomach:** Storage and mixing.
- **Small Intestine:** Absorb.
- **Large Intestine:** Water absorption and waste processing.

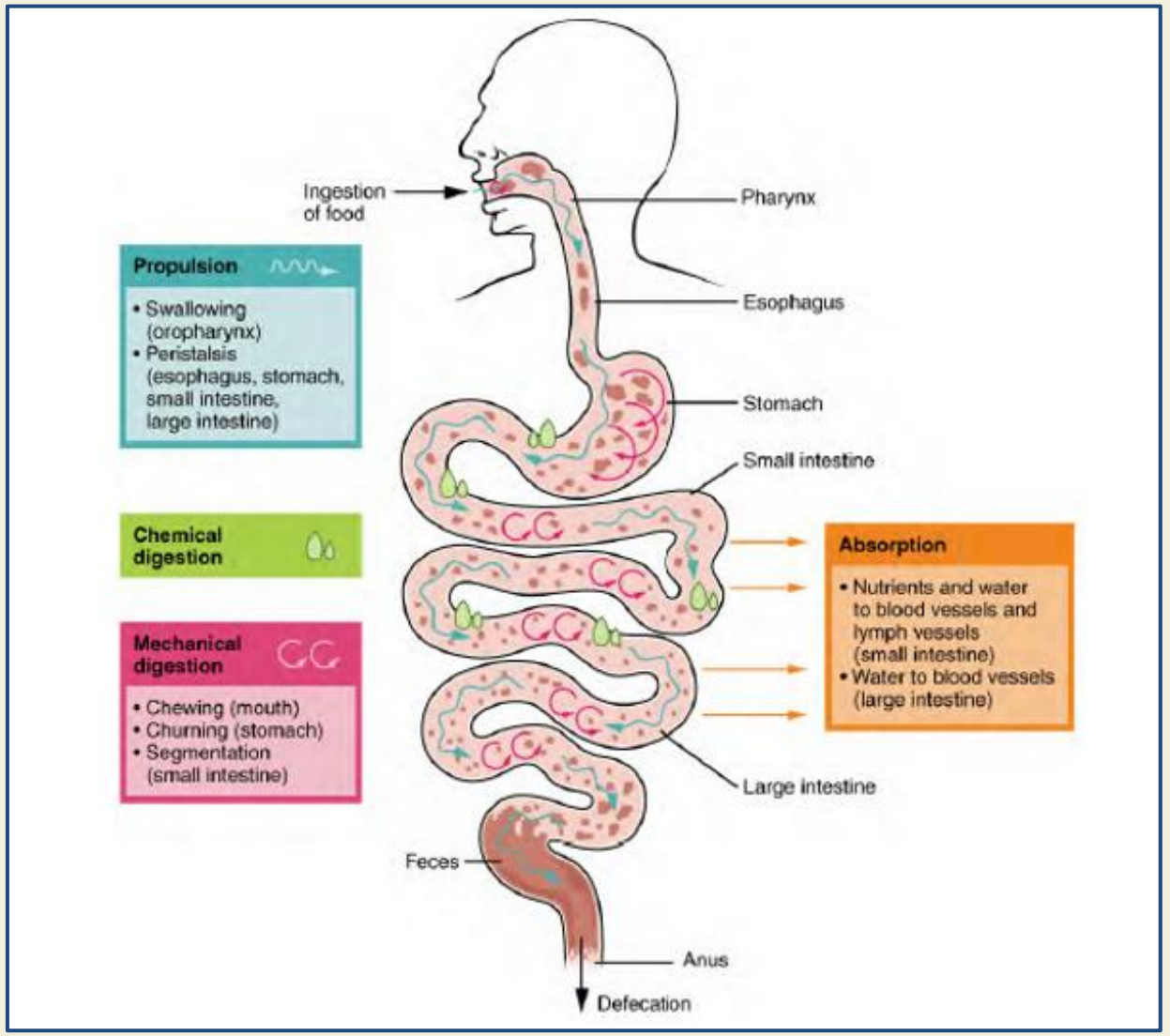
Accessory Organs

+ Salivary Glands, Liver, Gallbladder, Pancreas.

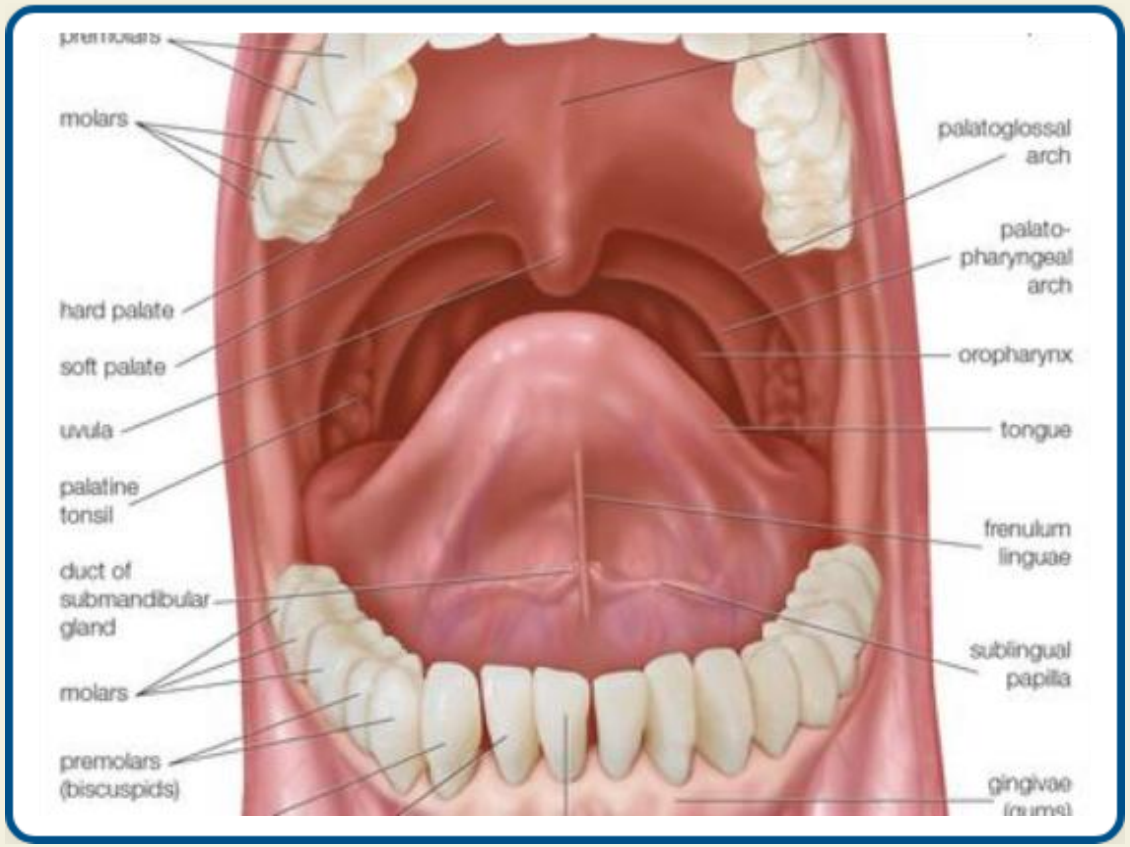


Activities of Digestion

- **Ingestion:** Taking food into the mouth.
- **Propulsion:** Moving food along (Swallowing & Peristalsis).
- **Mechanical Digestion:** Chewing and churning.
- **Chemical Digestion:** Enzymatic breakdown.
- **Absorption:** Nutrients entering blood or lymph.
- **Defecation:** Elimination of feces.



The Mouth



Initial Processing

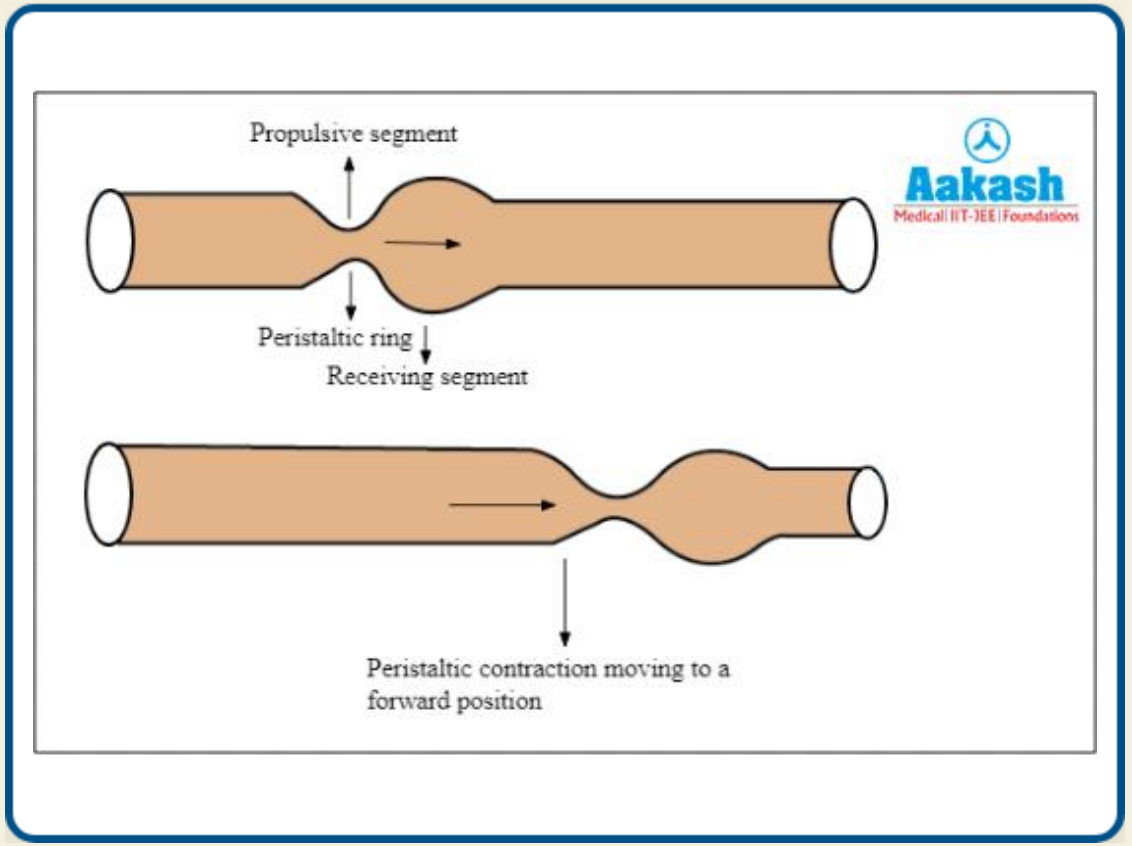
- **Mechanical:** Breaking down and mixing via teeth and tongue.
- **Chemical:** Salivary glands secrete amylase to initiate carbohydrate breakdown.
- **Propulsion:** Pushing the compacted bolus down to the pharynx.

The Esophagus

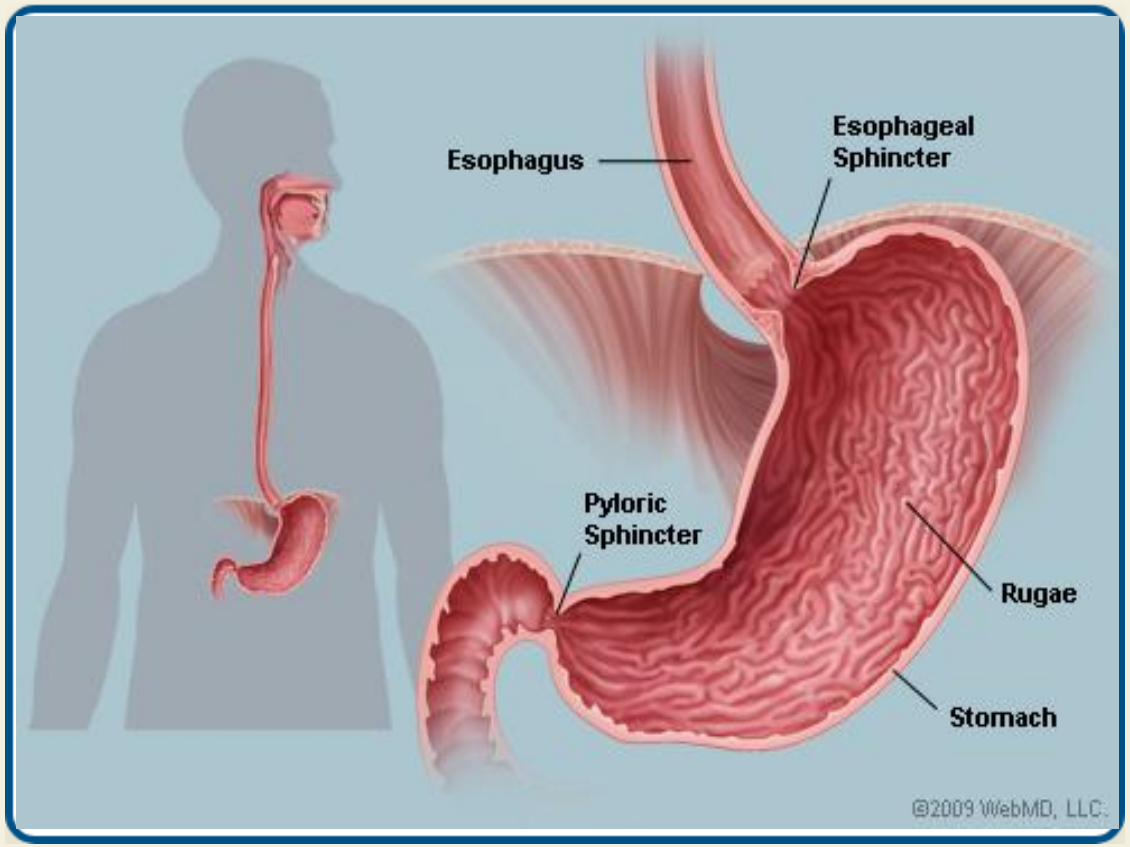
Peristalsis

Peristalsis is the rhythmic, wave-like contraction of smooth muscles along the gastrointestinal tract.

- Creates a propulsive wave that actively forces the bolus down into the stomach, independent of gravity.



The Stomach



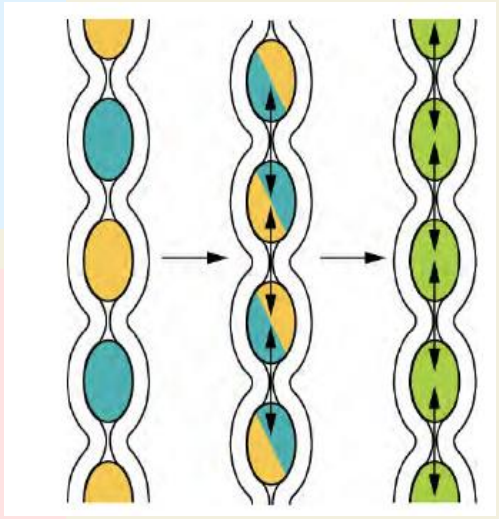
Storage & Churning

- **Digestion:** Powerful mechanical churning combined with HCl and enzymes forms acidic chyme.

Small & Large Intestine

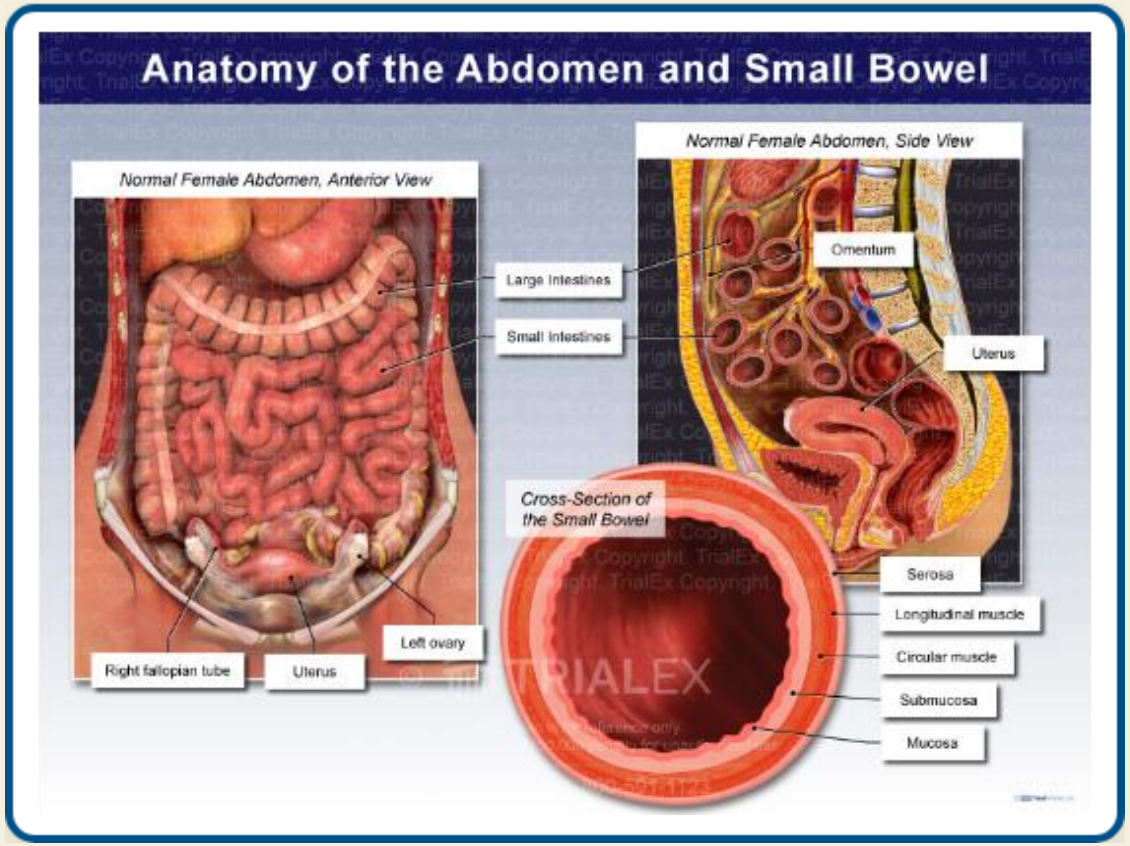
Small Intestine

Comprises Duodenum, Jejunum, and Ileum. The primary site where **90% of nutrient absorption** occurs.



Large Intestine

Absorbs remaining water, forms solid feces, and propels waste out through the rectum.

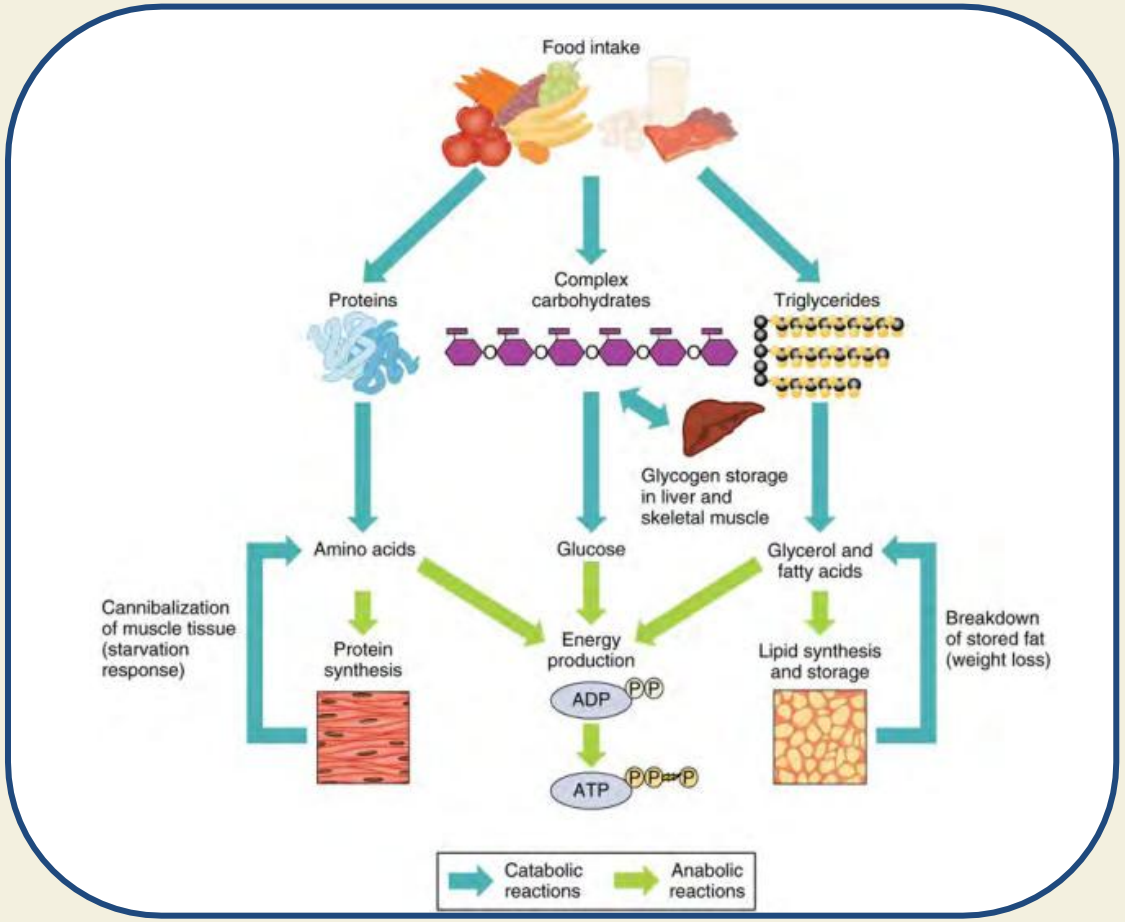


Metabolism & Nutrition

Cellular Energy

The ultimate goal of digestion is providing the raw materials for cellular metabolism.

- **Catabolism:** Breaking down carbohydrates, proteins, and lipids to generate energy (ATP).
- **Anabolism:** Building new structural components for the body using amino and fatty acids.



Human metabolism and nutrition pathways diagram: carbohydrates, proteins, lipids to ATP

System Interconnections

